

This SuKS

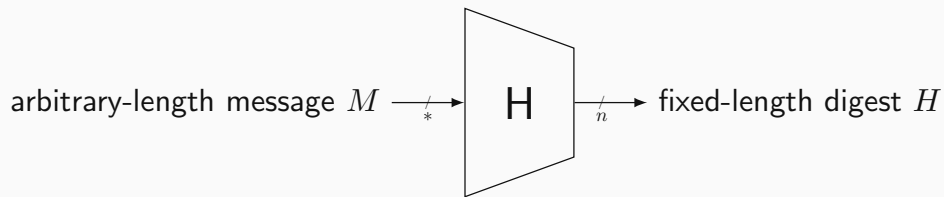
Bart Mennink

GAPS 2026 – Vienna

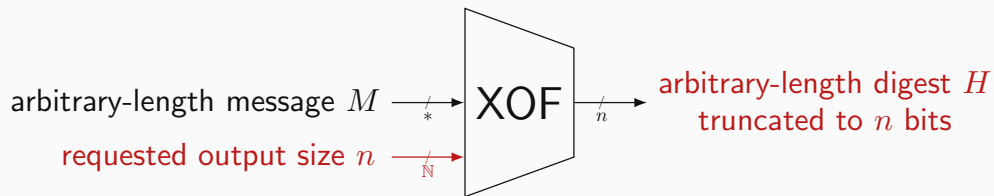
September 23, 2026

Introduction

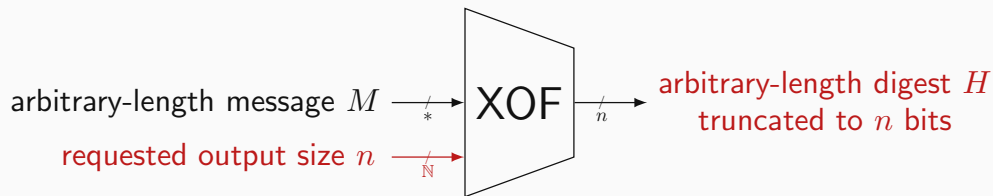
Cryptographic Hash Function



eXtendable Output Function

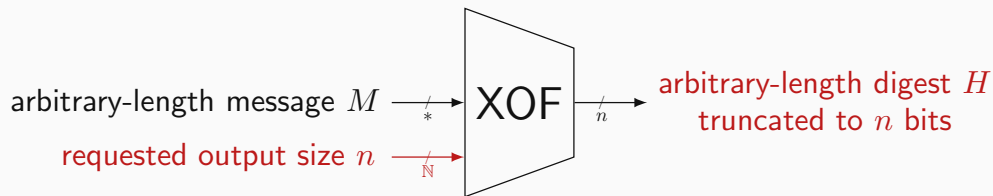


eXtendable Output Function



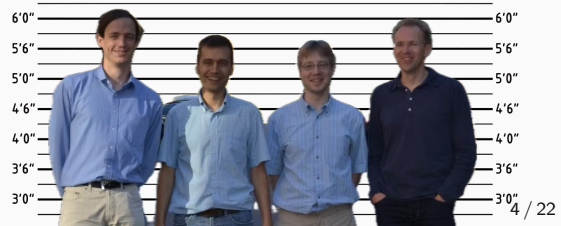
- Ideally behaves like a **random oracle RO** [BR93]

eXtendable Output Function

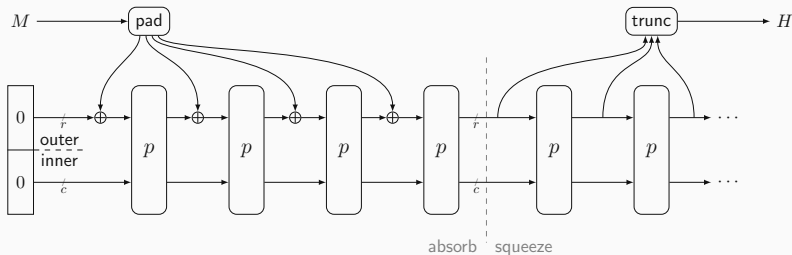


- Ideally behaves like a **random oracle RO** [BR93]
 - Different messages $\rightarrow$ independent random-looking digests
 - Same message but different output sizes $\rightarrow$ prefixed digests

Sponges [BDPV07, BDPV08]



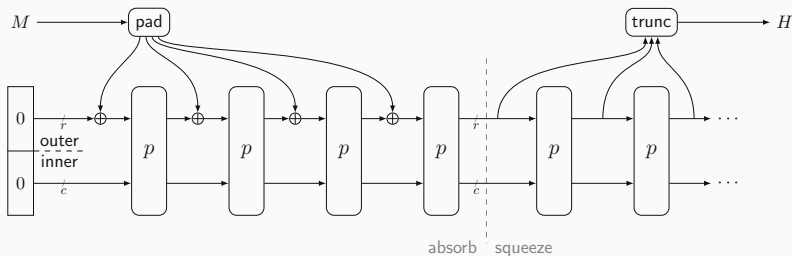
Sponges [BDPV07, BDPV08]



- p is a b -bit permutation, with $b = r + c$
 - r is the rate
 - c is the capacity (security parameter)



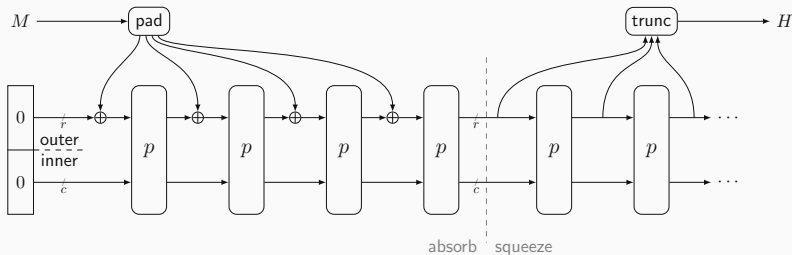
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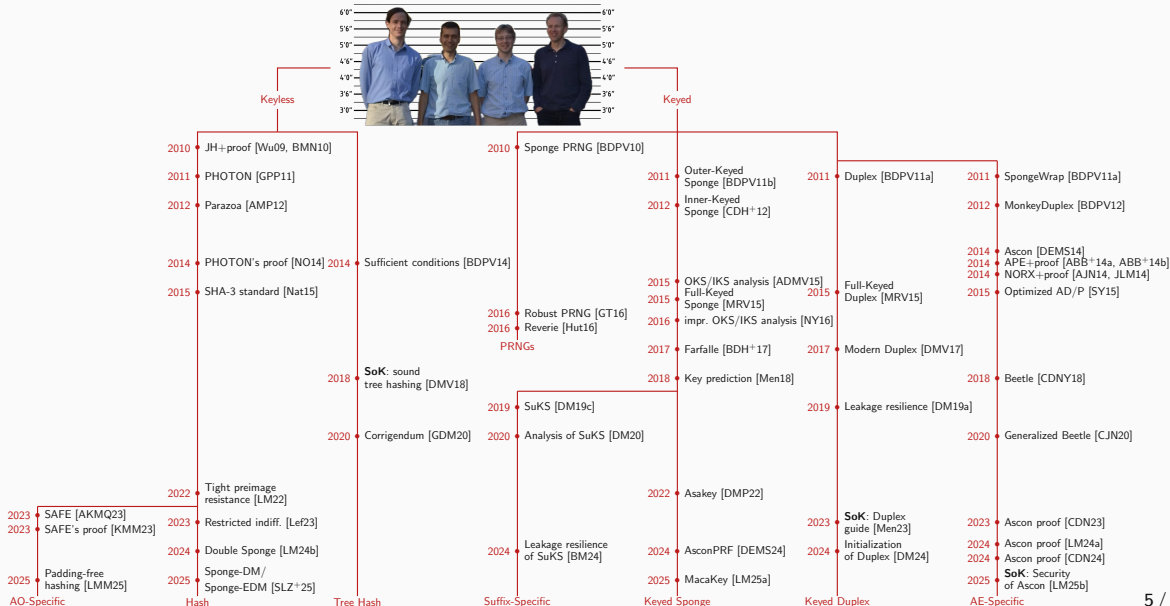
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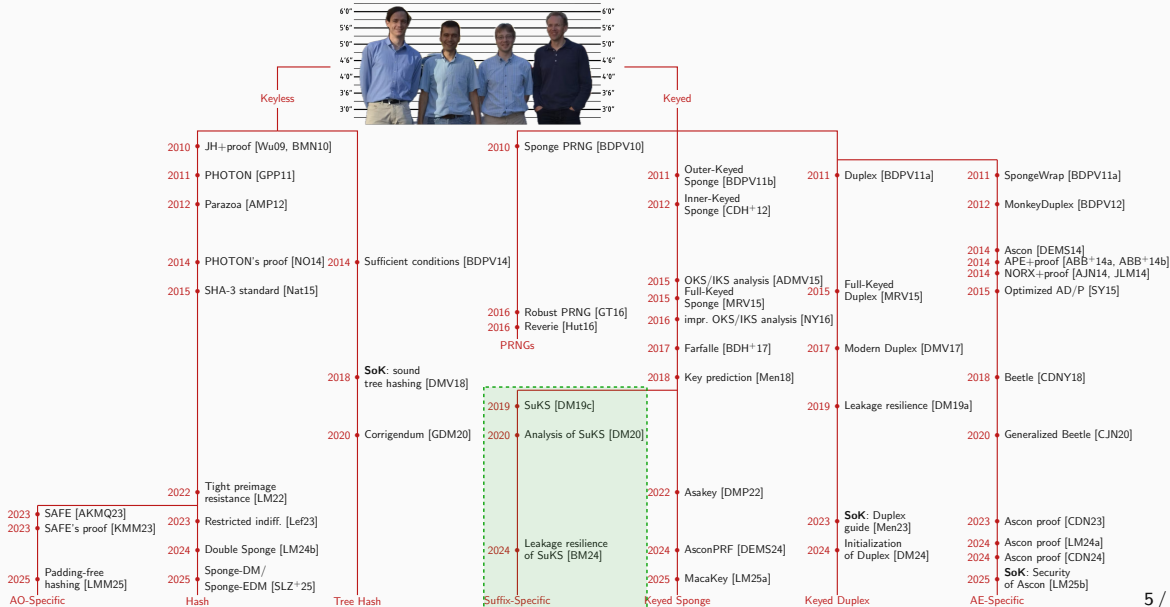
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 - r is the rate
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- Behaves like a RO up to $2^{c/2}$ queries
- Can be used in **keyed mode**: $\text{sponge}(K\|M)$



Non-Exhaustive History of Sponge Modes [Men26]

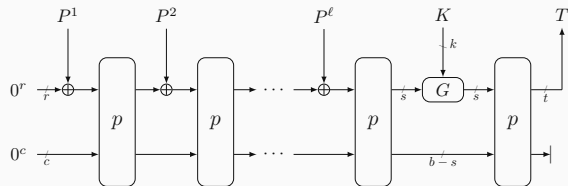


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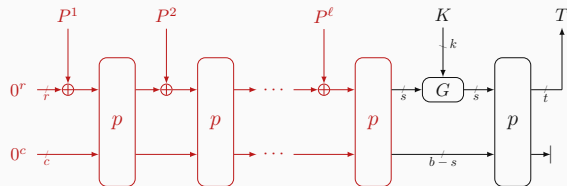


SuKS: Suffix Keyed Sponge

SuKS: Suffix Keyed Sponge [DM19c]

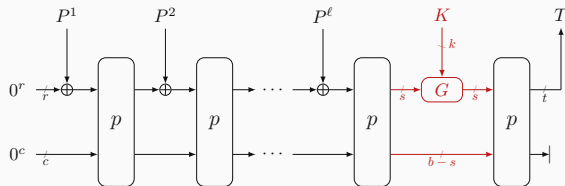


SuKS: Suffix Keyed Sponge [DM19c]



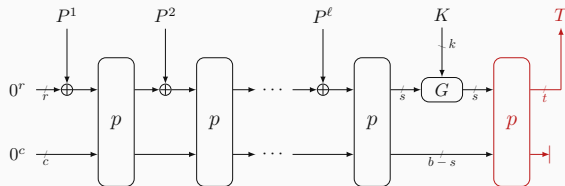
- 1 Absorbs data as in the keyless sponge

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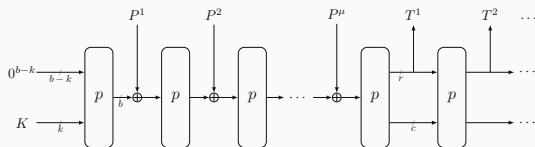
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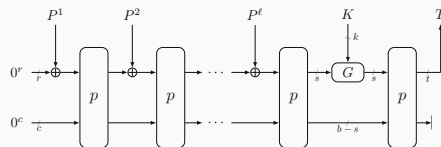


- 1 Absorbs data as in the keyless sponge
- 2 Absorbs key using some function G (a simple XOR suffices)
- 3 Squeezes t -bit tag after final permutation call

SuKS Versus Full-State Keyed Sponge

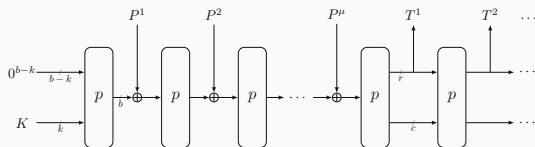


FSKS [BDPV12, MRV15]

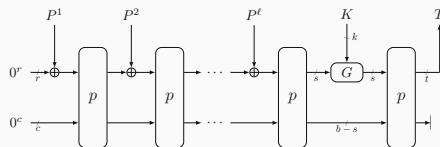


SuKS [DM19c]

SuKS Versus Full-State Keyed Sponge



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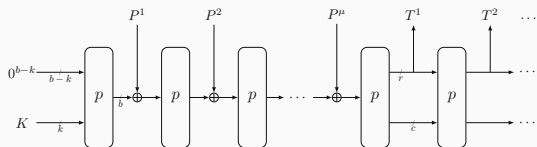


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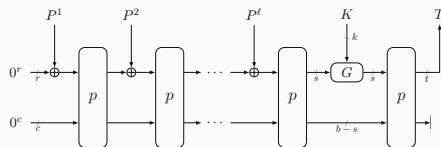
Disadvantages of SuKS

- No full-state absorption
- At best $c/2$ -bit security (offline attack)

SuKS Versus Full-State Keyed Sponge



FSKS [BDPV12, MRV15]



SuKS [DM19c]

Disadvantages of SuKS

- No full-state absorption
- At best $c/2$ -bit security (offline attack)

Advantages of SuKS

- Side-channel leakage limited
- $s, t \leq b$ arbitrary (typical: $s = t = c/2$)

Intermezzo: Multicollision Coefficient (1/2)

Definition

- M balls, 2^r bins
- $\mu_{r,c}^M$ is smallest x such that $\Pr(|\text{fullest bin}| > x) \leq \frac{x}{2^c}$

Intuition of Behavior

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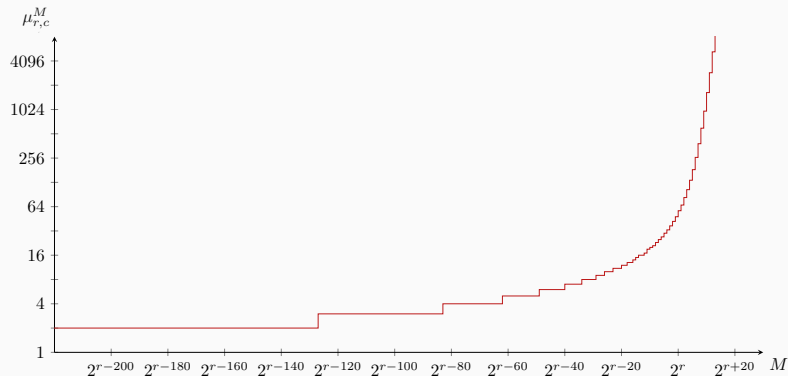
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Intuition of Behavior

- If $M \ll 2^r$, all bins will likely be “reasonably” empty
- If $M \gg 2^r$, there will likely be a bin with around $\text{linear}(b) \cdot \frac{M}{2^r}$ balls
- Formula for $\mu_{r,c}^M$, and upper bounds in above 2 cases, derived in [DMV17]
- $\mu_{r,c}^M$ is (at most) smallest x that satisfies

$$\frac{2^b e^{-M/2^r} (M/2^r)^x}{(x - M/2^r)x!} \leq 1$$

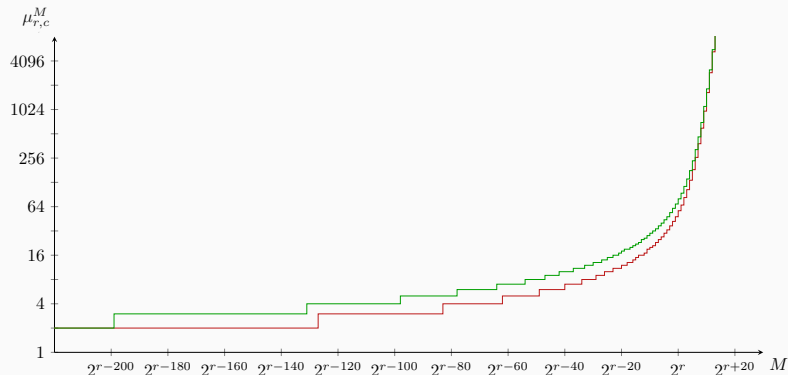
Stairway to Heaven for $b = 256$



$M/2^r$	$\mu_{r,c}^M$
2^{-256}	—
2^{-128}	2
2^{-64}	4
2^{-32}	8
2^{-16}	14
2^{-8}	23
2^0	57
2^8	601
2^{16}	70205
2^{19}	537313

Intermezzo: Multicollision Coefficient (2/2)

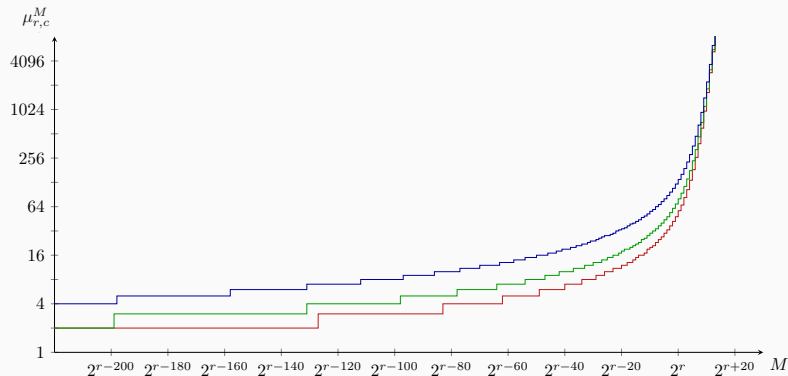
Stairway to Heaven for $b = 256$, $b = 400$



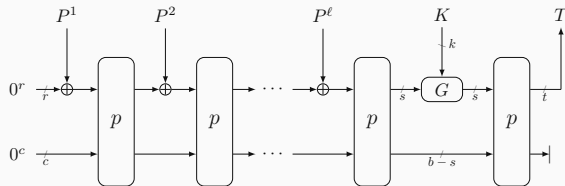
$M/2^r$	$\mu_{r,c}^M$	$\mu_{r,c}^M$
2^{-256}	—	2
2^{-128}	2	4
2^{-64}	4	7
2^{-32}	8	12
2^{-16}	14	21
2^{-8}	23	34
2^0	57	80
2^8	601	707
2^{16}	70205	71484
2^{19}	537313	540887

Intermezzo: Multicollision Coefficient (2/2)

Stairway to Heaven for $b = 256$, $b = 400$, $b = 800$

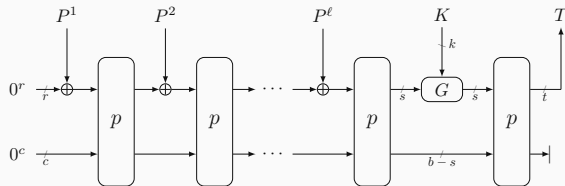


$M/2^r$	$\mu_{r,c}^M$	$\mu_{r,c}^M$	$\mu_{r,c}^M$
2^{-256}	—	2	4
2^{-128}	2	4	7
2^{-64}	4	7	12
2^{-32}	8	12	23
2^{-16}	14	21	40
2^{-8}	23	34	64
2^0	57	80	139
2^8	601	707	944
2^{16}	70205	71484	74119
2^{19}	537313	540887	548194



- $k, s, t \leq b$
- G is $2^{-\delta}$ -uniform and $2^{-\epsilon}$ -universal

$$\mathbf{Adv}_F^{\text{prf}}(\mathcal{A}) \leq \frac{2N^2}{2^c} + \frac{\mu_{b-s,s}^{2(N-q)} \cdot N}{2^{\min\{\delta,\epsilon\}}} + \frac{\mu_{t,b-t}^q \cdot N}{2^{b-t}}$$

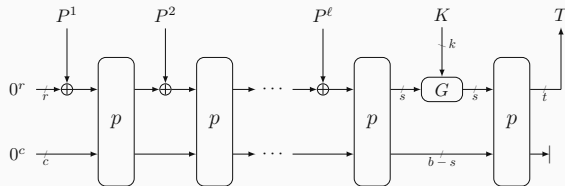


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inner collision

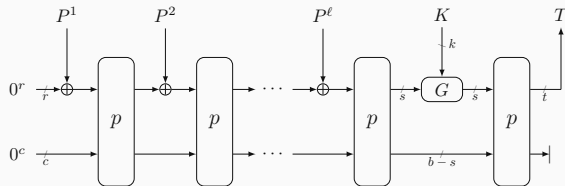
Security of SuKS [DM19c]



- $k, s, t \leq b$
- G is $2^{-\delta}$ -uniform and $2^{-\epsilon}$ -universal

$$\text{Adv}_F^{\text{prf}}(\mathcal{A}) \leq \underbrace{\frac{2N^2}{2^c}}_{\text{inner collision}} + \frac{\mu_{b-s,s}^{2(N-q)} \cdot N}{2^{\min\{\delta,\epsilon\}}} + \underbrace{\frac{\mu_{t,b-t}^q \cdot N}{2^{b-t}}}_{\text{"break at } T", \text{ bounds construction queries with same tag}}$$

Security of SuKS [DM19c]



- $k, s, t \leq b$
- G is $2^{-\delta}$ -uniform and $2^{-\epsilon}$ -universal

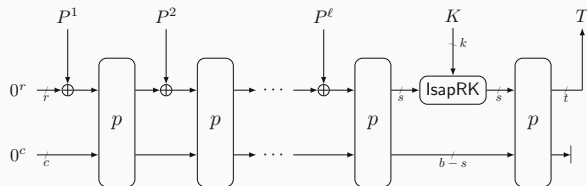
$$\text{Adv}_F^{\text{prf}}(\mathcal{A}) \leq \frac{2N^2}{2^c} + \frac{\mu_{b-s,s}^{2(N-q)} \cdot N}{2^{\min\{\delta,\epsilon\}}} + \frac{\mu_{t,b-t}^q \cdot N}{2^{b-t}}$$

inner collision

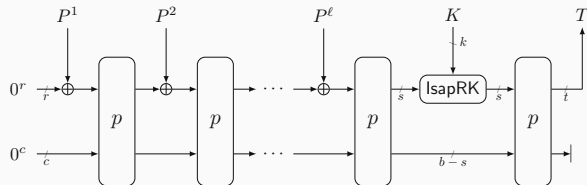
“break at G ”, bounds primitive queries with same inner part

“break at T ”, bounds construction queries with same tag

Application to MAC Part of ISAP [DEM⁺20, DEM⁺19]



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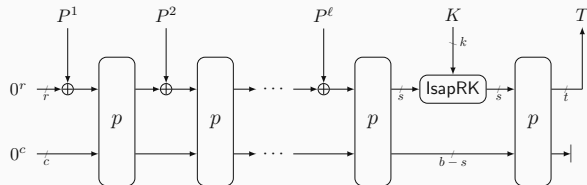


$$(b, c, r, k) = (400, 256, 144, 128)$$

- $\mu_{b-s,s}^{2(N-q)} = \mu_{272,128}^{2^{129}} \leq 3$
- $\mu_{t,b-t}^q = \mu_{128,272}^{2^{128}} \leq 80$

$$\mathbf{Adv}_{\text{IsapMAC}}^{\text{prf}}(\mathcal{A}) \leq \frac{2N^2}{2^{256}} + \frac{3N}{2^{128}} + \frac{80N}{2^{272}}$$

Application to MAC Part of ISAP [DEM⁺20, DEM⁺19]



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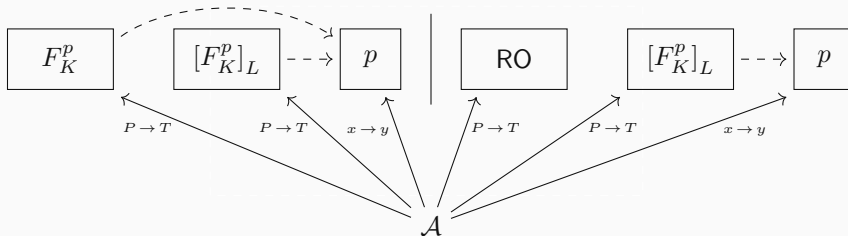
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$$(b, c, r, k) = (320, 256, 64, 128)$$

- $\mu_{b-s,s}^{2(N-q)} = \mu_{192,128}^{2^{129}} \leq 5$
- $\mu_{t,b-t}^q = \mu_{128,192}^{2^{128}} \leq 67$

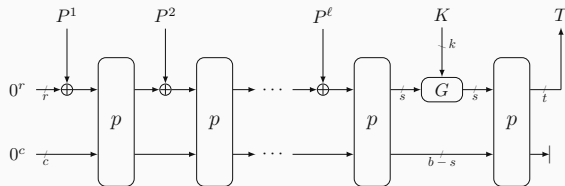
$$\text{Adv}_{\text{IsapMAC}}^{\text{prf}}(\mathcal{A}) \leq \frac{2N^2}{2^{256}} + \frac{5N}{2^{128}} + \frac{67N}{2^{192}}$$



$$\text{Adv}_F^{\text{nalr-prf}}(\mathcal{A}) = \max_{L \in \mathcal{L}} \Delta_{\mathcal{A}}([F_K^p]_L, F_K^p, p; [F_K^p]_L, RO_{*,t}, p)$$

- Non-adaptive leakage resilience
- Bounded leakage model

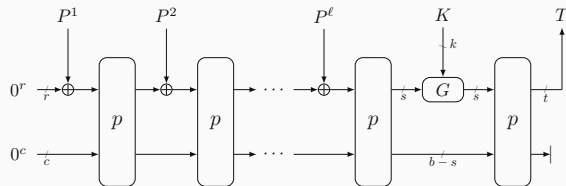
Leakage Resilience of SuKS



- $k, s, t \leq b$
- G is strongly protected, $2^{-\delta}$ -uniform, and $2^{-\epsilon}$ -universal

$$\mathbf{Adv}_F^{\text{nalr-prf}}(\mathcal{A}) \leq \frac{2N^2}{2^c} + \frac{\mu_{s,b-s}^{2(N-q)}}{2^{b-s}} + \frac{\mu_{b-s,s}^{2(N-q)} \cdot N}{2^{\min\{\delta,\epsilon\} - \mu_{s,b-s}^{2(N-q)} \lambda}} + \frac{\mu_{t,b-t}^{2q} \cdot N}{2^{b-t-\lambda}}$$

Leakage Resilience of SuKS

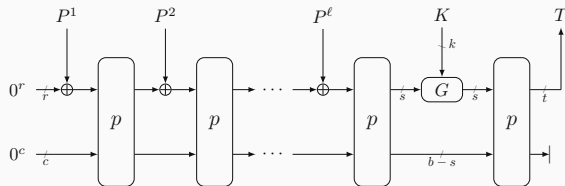


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bounds the number of repeated leakages on same $G(K, X)$

Leakage Resilience of SuKS



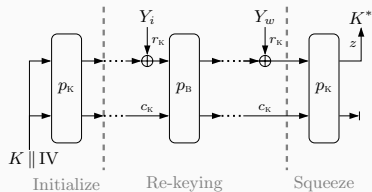
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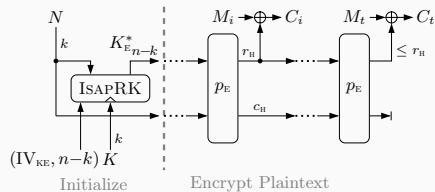
bounds the number of repeated leakages on same $G(K, X)$

bounds leakage from
secret next to T

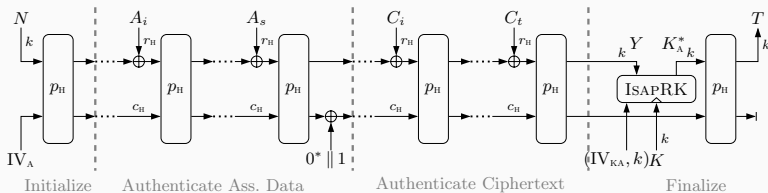
Leakage Resilience of ISAP Mode [DM19b]



IsapRK

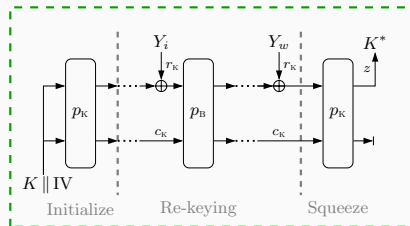


IsapEnc



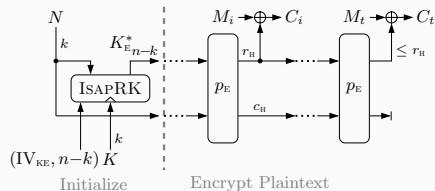
IsapMAC

Leakage Resilience of ISAP Mode [DM19b]

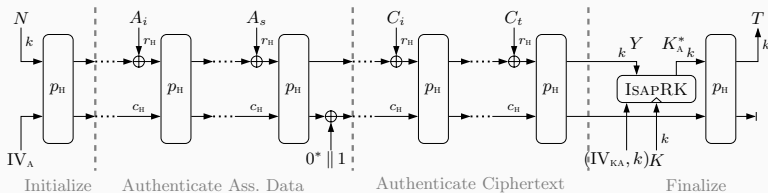


IsapRK

Keyed duplex
with small rate

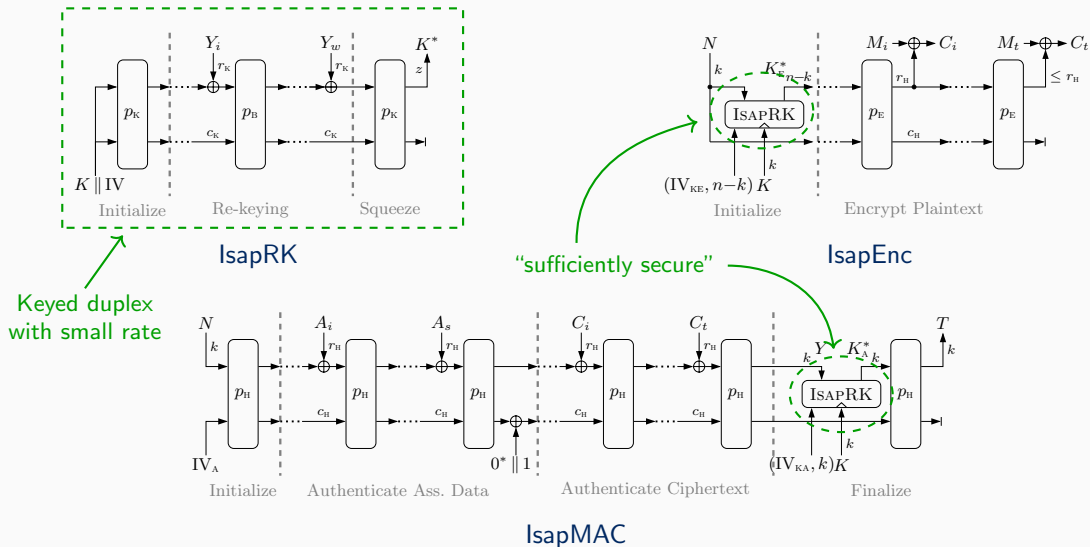


IsapEnc

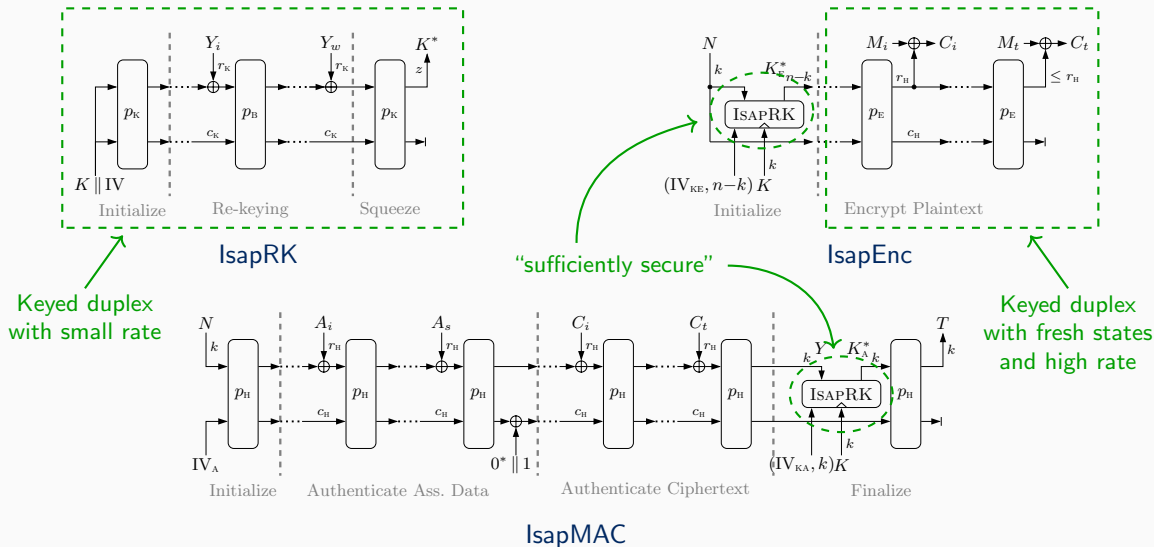


IsapMAC

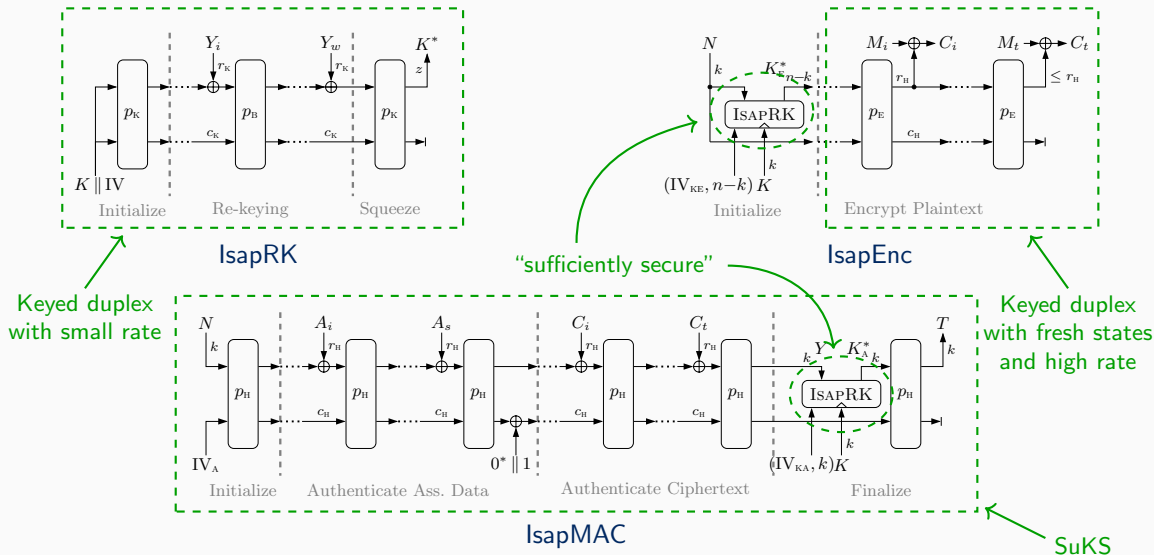
Leakage Resilience of ISAP Mode [DM19b]



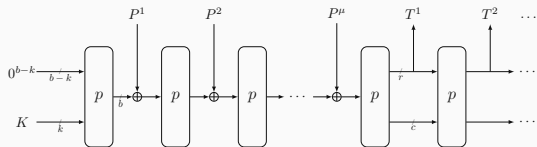
Leakage Resilience of ISAP Mode [DM19b]



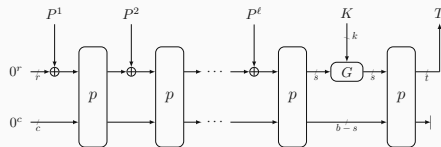
Leakage Resilience of ISAP Mode [DM19b]



Limitations of SuKS

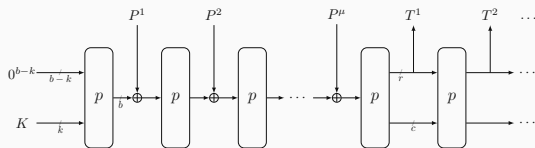


FSKS

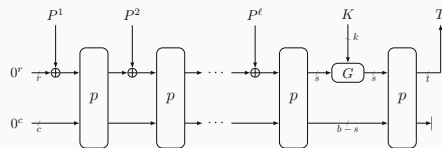


SuKS

Limitations of SuKS



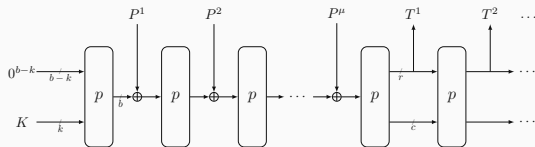
FSKS



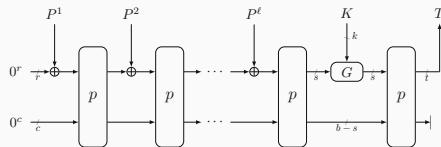
SuKS

SuKS Does Not Use the Sponge to the Fullest

Limitations of SuKS



FSKS

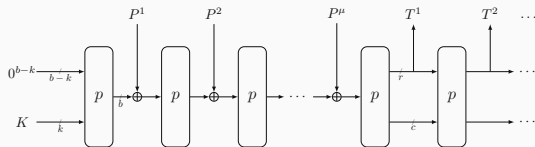


SuKS

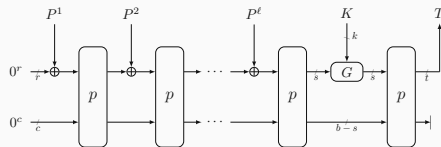
SuKS Does Not Use the Sponge to the Fullest

- Full-state absorption?
- Arbitrary-length squeezing?

Limitations of SuKS



FSKS



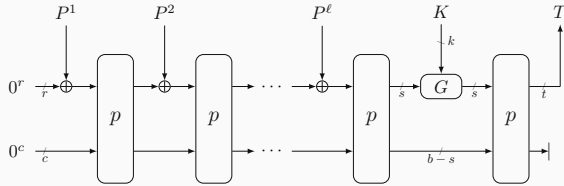
SuKS

SuKS Does Not Use the Sponge to the Fullest

- Full-state absorption?
- Arbitrary-length squeezing?
- Multi-user security analysis?

SuKSessor: SuKS's Successor

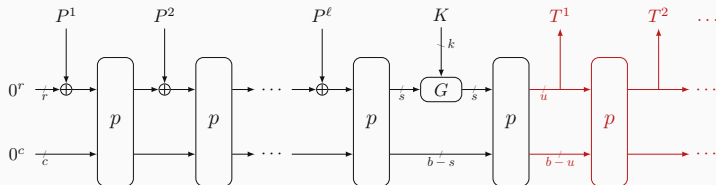
SuKSession: Suffix Keyed Sponge Error



Observation 1

Observation 2

SuKSession: Suffix Keyed Sponge Error

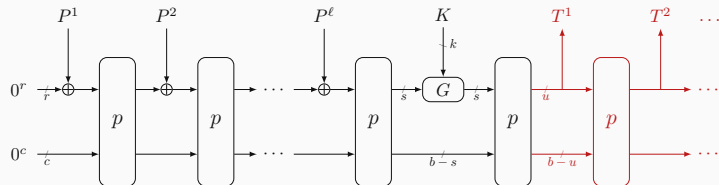


Observation 1

- FSKS allows for arbitrary-length squeezing
- There is no reason to restrict SuKS to a single squeeze only

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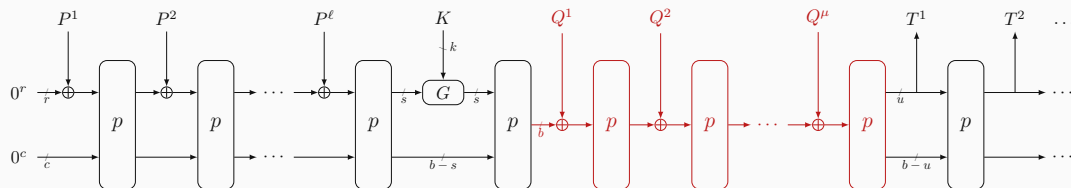
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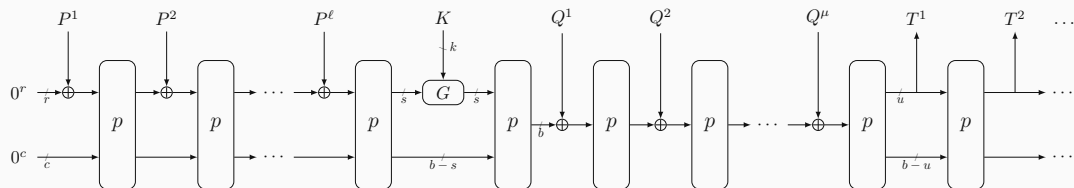
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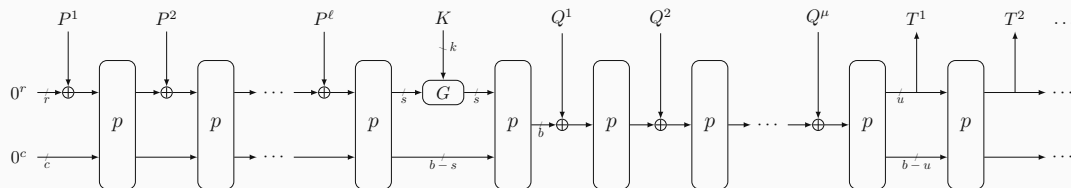
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SuKSessor: Suffix Keyed Sponge Error



**We prove black-box security and leakage resilience
of SuKSessor in the multi-user setting**

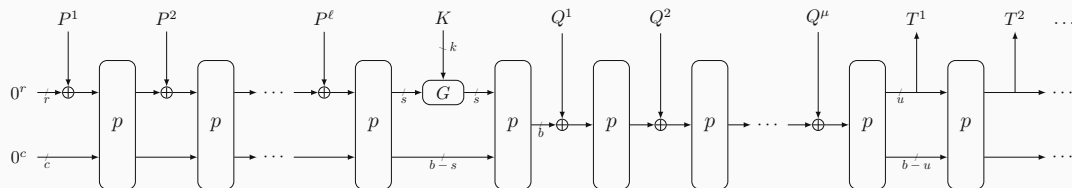
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**We prove black-box security and leakage resilience
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- We require **uniqueness of P** (as in the original SuKS). . .
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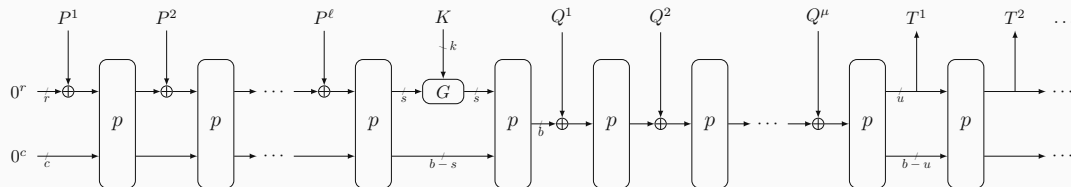
SuKSessor: Suffix Keyed Sponge Eссор



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- For empty Q and $|T| \leq u$: original SuKS retained

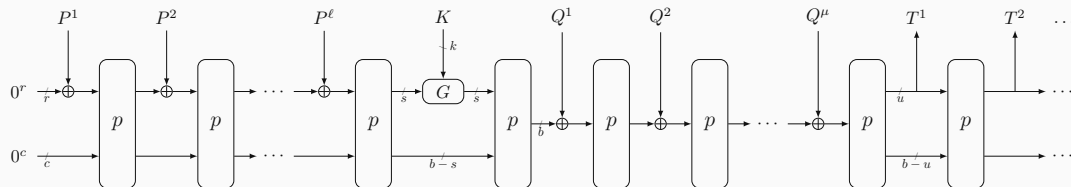
Multi-User Security of SuKSessor (1/2)



- $k, s, u \leq b$
- G is $2^{-\delta}$ -uniform and $2^{-\epsilon}$ -universal
- $q_{\max}$ is maximum number of users evaluating a single P

$$\text{Adv}_F^{\text{mu-prf}}(\mathcal{A}) \leq \frac{2N^2}{2^c} + \frac{q_{\max} \cdot \mu_{b-s,s}^{2(N-\sigma)} \cdot N}{2^{\min\{\delta,\epsilon\}}} + \frac{\mu_{u,b-u}^{\sigma_{\text{sq}}} \cdot N}{2^{b-u}}$$

Multi-User Security of SuKSessor (1/2)

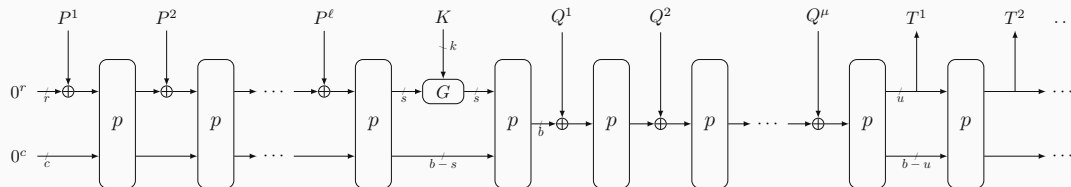


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extension
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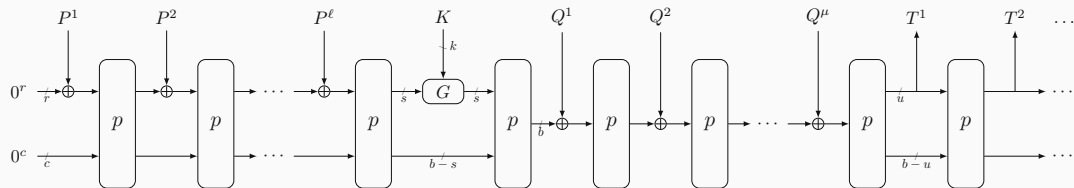
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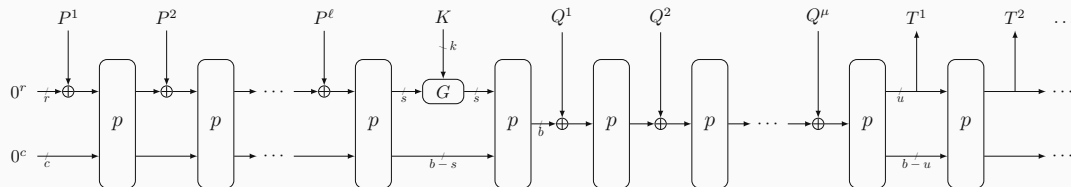
$$\text{Adv}_F^{\text{mu-prf}}(\mathcal{A}) \leq \frac{2N^2}{2^c} + \underbrace{q_{\max}}_{\text{only "true" multi-user security loss}} \cdot \frac{\mu_{b-s,s}^{2(N-\sigma)} \cdot N}{2^{\min\{\delta, \epsilon\}}} + \underbrace{\frac{\mu_{u,b-u}^{\sigma_{\text{sq}}} \cdot N}{2^{b-u}}}_{\text{extension to multiple squeezes}}$$

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Interpretation and Remarks

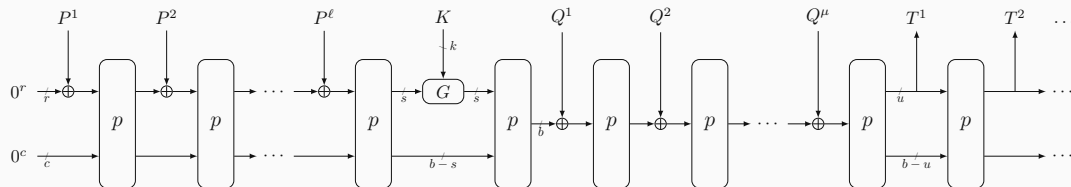
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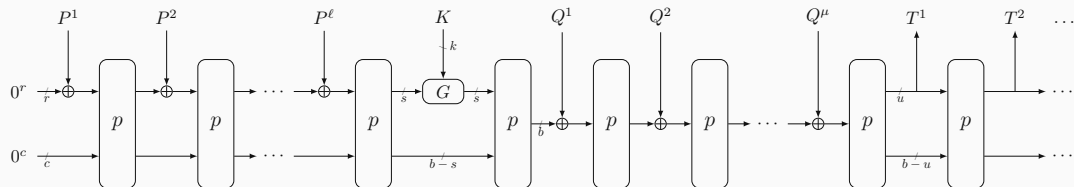
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- For 1 user, empty Q , and $|T| \leq u$, bound **identical** to that of SuKS
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- Squeezing up to roughly $u \approx r + c/2 - \log_2(c)$ bits, **per permutation**

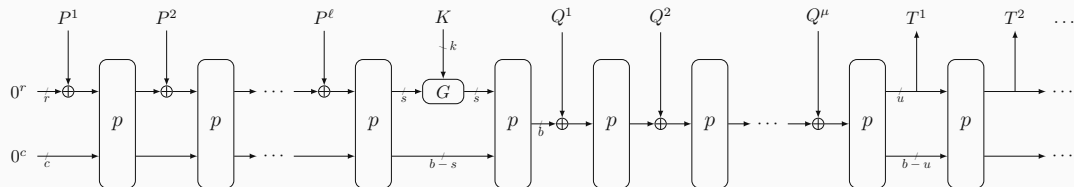
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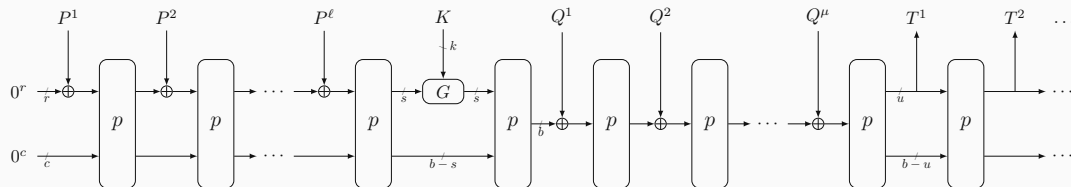
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- All terms are matched with a generic attack [DM20]

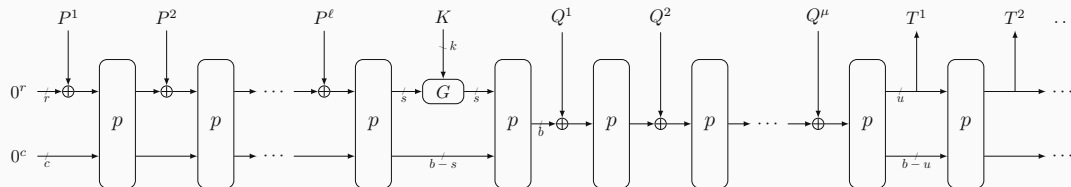
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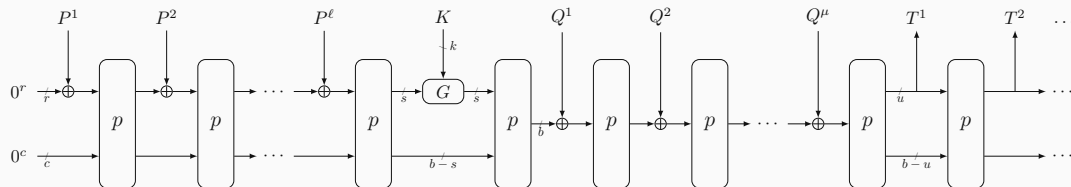


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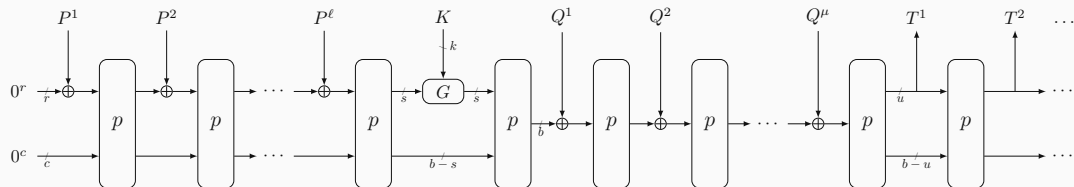
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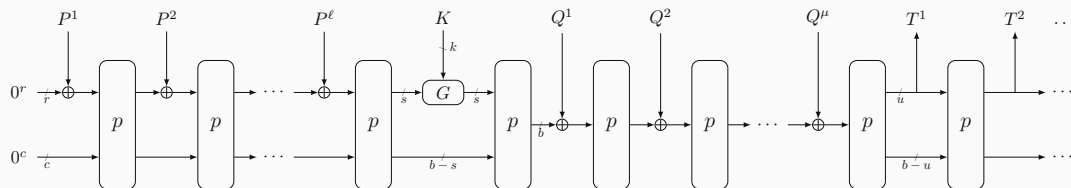
extra term (was hidden in $\frac{2N^2}{2^c}$ before)
 bounds extra leakage next to any T^i

Parameter Interpretation



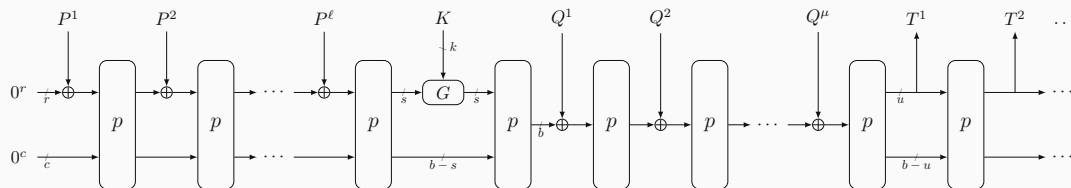
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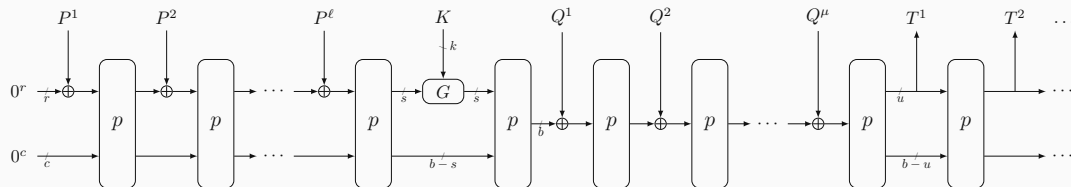
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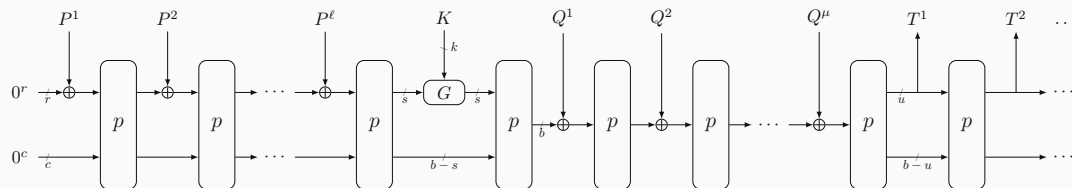
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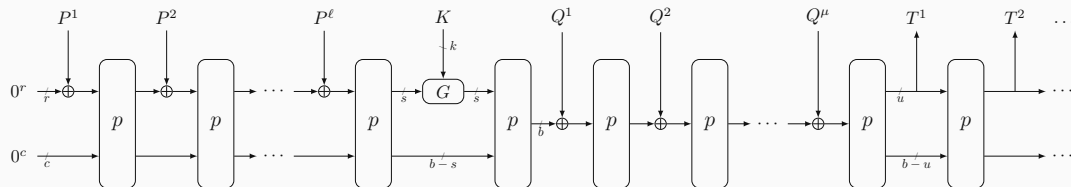


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≤ 5

u	64	128	176	184	192
$\mu_{u,320-u}^{2^{64}}$	67	5	3	3	3
$\frac{\mu_{u,320-u}^{2^{64}} \cdot N}{2^{320-u}}$	$2^{-121.9}$	$2^{-61.7}$	$2^{-14.4}$	$2^{-6.4}$	3

Conclusion

SuKSessor

- SuKS: easy-to-protect message authentication as used in ISAP v2
- SuKSessor generalizes SuKS at almost no cost to:
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Thank you!

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